

SPECTRA OF MULTIPLICATION OPERATORS AS A NUMERICAL TOOL*

B. VIOREANU[†] AND V. ROKHLIN[‡]

Abstract. We introduce a numerical procedure for the construction of interpolation and quadrature formulae on bounded convex regions in the plane. The construction is based on the behavior of spectra of certain multiplication operators and leads to nodes which are inside a prescribed convex region in $\mathbb{R}^2$. The resulting interpolation schemes are numerically stable and the quadrature formulae have positive weights and almost (but not quite) optimal numbers of nodes. The performance of the algorithm is illustrated by several numerical examples.

Key words. multiplication operator, interpolation, numerical integration

AMS subject classifications. 65D05, 65D30, 65D32, 65F15

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1. Introduction. Numerical integration and interpolation constitute one of the oldest areas of numerical analysis and find applications in most branches of applied mathematics. In general outline, the formulation is as follows. Given a region $\Omega \subset \mathbb{R}^k$ (usually with $k \leq 3$) and a collection of functions $\varphi_1, \varphi_2, \dots, \varphi_n$ mapping Ω to $\mathbb{R}^1$, we would like to find a collection of points $x_1, x_2, \dots, x_m$ with the property that for each point x in Ω , there exist coefficients $\alpha_1(x), \alpha_2(x), \dots, \alpha_m(x)$ such that

$$(1.1) \quad f(x) = \sum_{j=1}^m \alpha_j(x) f(x_j)$$

for all $f : \Omega \rightarrow \mathbb{R}^1$ of the form

$$(1.2) \quad f(x) = \sum_{i=1}^n \beta_i \varphi_i(x);$$

in other words, the interpolation formula (1.1) is exact for any f that is a linear combination of the functions $\{\varphi_i\}$.

In practice, (1.1) is not sufficient, since if the resulting interpolation formulae are to be of any practical use, the coefficients $\{\alpha_i(x)\}$ must not be very large and have to be easy to evaluate. The following theorem (see [11]) guarantees the existence of numerically acceptable interpolation coefficients under extremely weak conditions.

THEOREM 1.1. *Suppose S is an arbitrary set, n is a positive integer, $f_1, f_2, \dots, f_n$ are bounded complex-valued functions on S , and ε is a positive real number less than or equal to 1. Then, there exist n points $x_1, x_2, \dots, x_n$ in S and n functions $\alpha_1(x), \alpha_2(x), \dots, \alpha_n(x)$ on S such that*

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$$(1.3) \quad |\alpha_j(x)| \leq 1 + \varepsilon$$

for all x in S and $j = 1, 2, \dots, n$, and

$$(1.4) \quad f(x) = \sum_{j=1}^n \alpha_j(x) f(x_j)$$

for all x in S and any function f defined on S via the formula

$$(1.5) \quad f(x) = \sum_{i=1}^n \beta_i f_i(x)$$

for some complex numbers $\beta_1, \beta_2, \dots, \beta_n$.

Thus, the *existence* of a stable interpolation formula is never a serious issue; this paper deals with numerical procedures for constructing such interpolation formulae.

Numerical integration of functions is closely related to their numerical interpolation but involves a number of subtleties. Given a region $\Omega \in \mathbb{R}^k$ and a collection of functions $\varphi_1, \varphi_2, \dots, \varphi_n : \Omega \rightarrow \mathbb{R}^1$, one wishes to find points $x_1, x_2, \dots, x_m \in \Omega$ and coefficients $w_1, w_2, \dots, w_m \in \mathbb{R}^1$ such that for each of the functions $\{\varphi_i\}$,

$$(1.6) \quad \int_{\Omega} \varphi_i(x) dx = \sum_{j=1}^m w_j \varphi_i(x_j),$$

the pair $\{\{x_i\}; \{w_i\}\}$ is frequently referred to as a quadrature formula for the functions $\{\varphi_i\}$ with $\{x_i\}$ the nodes and $\{w_i\}$ the weights of the quadrature.

Again, (1.6) is not sufficient for the quadrature $\{\{x_i\}, \{w_i\}\}$ to be of practical use. Indeed, its numerical stability will be poor if the absolute values of the weights $\{w_i\}$ are large. Obviously, if all the weights $\{w_i\}$ are positive, this issue does not arise, and quadratures with positive weights are generally preferred (though in most cases not absolutely necessary).

Obviously, given a stable interpolation formula of the form (1.4) for a set of functions $f_1, f_2, \dots, f_n$, it is easy to construct a quadrature that is numerically stable; this observation is formalized in the following corollary to Theorem 1.1 (see [11]).

COROLLARY 1.2. *Suppose S is a measure space, w is a nonnegative real-valued integrable function on S (that serves as the weight for integration), n is a positive integer, $f_1, f_2, \dots, f_n$ are bounded complex-valued integrable functions on S , and $\varepsilon \leq 1$ is a positive real number.*

Then, there exist n complex numbers $w_1, w_2, \dots, w_n$ such that

$$(1.7) \quad |w_k| \leq (1 + \varepsilon) \int w(x) dx$$

for all $k = 1, 2, \dots, n$ and

$$(1.8) \quad \int f(x) w(x) dx = \sum_{k=1}^n w_k f(x_k)$$

for any function f defined on S via the formula (1.5), where $x_1, x_2, \dots, x_n$ are the n points in S chosen in Theorem 1.1.

One fundamental difference between interpolation and integration is the fact that interpolation in an n -dimensional space always requires at least n points (and, according to Theorem 1.1, n points are always sufficient); on the other hand, a quadrature

formula for integrating n functions often requires fewer than n points. The reader is referred to section 2.7 for a somewhat more detailed discussion of this class of issues.

In one dimension, the existing theory of quadrature formulae is quite satisfactory. *Gaussian quadratures* (see, for instance, [4]) are optimal for integrating polynomials up to a fixed degree, while *generalized Gaussian quadratures* are used for integrating more general classes of functions, including smooth functions and functions with endpoint singularities [10, 14].

In dimensions higher than one, much less is known regarding optimal numerical integration formulae. Classical expositions on constructing quadratures for multidimensional integrals can be found in [5], [8], [9], and a more recent overview of the development of the theory is given in [3].

This paper is partly devoted to constructing quadratures on convex regions in two dimensions exhibiting Gaussian behavior (the number of nodes of the quadrature is smaller than the number of functions to be integrated) using a simple scheme, which is relatively easy to implement. The quadratures have positive weights and all their nodes are interior to the region of integration. In addition, the quadratures are invariant under all or, in some cases, most of the symmetries of the region, if such symmetries are present.

The structure of the paper is as follows. Section 2 introduces mathematical and numerical preliminaries. In section 3 we develop the mathematical apparatus which is used by the algorithm described in section 4. The numerical results are summarized in section 5. Finally, section 6 contains conclusions.

2. Mathematical and numerical preliminaries.

2.1. Analytical preliminaries. In this section, we summarize a few well-known facts in functional analysis, which are used in the proof of Theorem 3.4.

Let X be a measurable subset of the complex plane. The linear space of square integrable complex-valued functions on X is generally denoted by $L^2(X)$. As is well known, $L^2(X)$ has the structure of a complex Hilbert space with the inner product defined by

$$(2.1) \quad (f, g) = \iint_X f(x, y) \overline{g(x, y)} dx dy.$$

Let U be an arbitrary finite-dimensional subspace of $L^2(X)$. We denote by M_{x+iy} the multiplication operator by $g(x, y) = x + iy$, defined on the space U :

$$(2.2a) \quad M_{x+iy} : U \longrightarrow L^2(X),$$

$$(2.2b) \quad M_{x+iy}(f(x, y)) = (x + iy) \cdot f(x, y).$$

The following theorem lists a few basic properties of the orthogonal projection operator. A proof can be found, for instance, in [2].

THEOREM 2.1. *Let X be a measurable subset of $\mathbb{C}$ and U be a nonzero finite-dimensional subspace of $L^2(X)$. The orthogonal projection P_U onto the subspace U has the following properties:*

1. P_U is a linear operator with operator norm equal to 1.
2. The restriction of P_U to the space U is the identity operator on U , i.e., $P_U(\varphi) = \varphi$ for all functions φ in U .
3. The operator P_U is self-adjoint, i.e., for all φ, ψ in $L^2(X)$,

$$(2.3) \quad (P_U(\varphi), \psi) = (\varphi, P_U(\psi)).$$

Under the conditions of Theorem 2.1, $P_U \circ M_{x+iy}$ is a linear operator on the finite-dimensional space U . Given an orthonormal basis $\{\varphi_1, \varphi_2, \dots, \varphi_n\}$ of U , we can find an explicit formula of the matrix A representing the operator $P_U \circ M_{x+iy}$ in this basis. Indeed, the entries (a_{ij}) of the matrix A are defined by

$$(2.4) \quad a_{ij} = ((P_U \circ M_{x+iy})(\varphi_j), \varphi_i).$$

By Theorem 2.1, the orthogonal projection operator is self-adjoint; hence,

$$(2.5) \quad a_{ij} = (M_{x+iy}(\varphi_j), P_U(\varphi_i)).$$

Since φ_i belongs to U , $P_U(\varphi_i) = \varphi_i$ for all $1 \leq i \leq n$ by Theorem 2.1. It follows that (2.5) is equivalent to

$$(2.6) \quad a_{ij} = (M_{x+iy}(\varphi_j), \varphi_i).$$

Finally, we can use the definitions of M_{x+iy} and of the inner product to express the right-hand side of (2.6) in the following explicit form:

$$(2.7) \quad a_{ij} = (M_{x+iy}(\varphi_j), \varphi_i)$$

$$(2.8) \quad = ((x + iy)\varphi_j(x, y), \varphi_i(x, y))$$

$$(2.9) \quad = \iint_X (x + iy) \cdot \varphi_j(x, y) \cdot \overline{\varphi_i(x, y)} \, dx \, dy.$$

In the algorithm described in section 4, (2.9) is used for finding a matrix representation of the operator $P_U \circ M_{x+iy}$; the matrix representation is then used to find the eigenvalues of $P_U \circ M_{x+iy}$.

2.2. Geometrical preliminaries. In this section, we state an easy consequence of the two-dimensional version of the hyperplane separation theorem. A proof can be found, for instance, in [1].

LEMMA 2.2. *Let X be a closed convex set in the plane and $\lambda \in \mathbb{C}$ be a point outside X . Then there exists a line L that strictly separates λ from X . In other words, X is completely contained in one of the open half-planes determined by L , while λ lies in the other open half-plane determined by L .*

Lemma 2.2 is used in the proof of Lemma 3.3 in section 3.

2.3. Interpolation matrix. Given a region $X \subset \mathbb{R}^2$ and a collection of real-valued functions $\varphi_1, \varphi_2, \dots, \varphi_N$ defined on X , a set of N points $x_1, x_2, \dots, x_N$ inside X yields an interpolation scheme on X . Let the matrix $B = (b_{ij})$ be defined by the equations

$$(2.10) \quad b_{ij} = \varphi_i(x_j), \quad i, j = 1, 2, \dots, N.$$

If the determinant of B is nonzero, B will be referred to as the *interpolation matrix* associated to the interpolation scheme. Throughout the paper, the condition number of the interpolation matrix is used as a measure of the stability of the interpolation scheme—the interpolation scheme is stable if the condition number of the associated interpolation matrix is small.

2.4. Singular value decomposition. The singular value decomposition (SVD) is a ubiquitous tool in numerical analysis. In the case of real matrices, the existence of the SVD is guaranteed by the following lemma (see, for instance, [7] for more details).

LEMMA 2.3. *For any $n \times m$ real matrix A , there exist a positive integer p , an $n \times p$ real matrix U with orthonormal columns, an $m \times p$ real matrix V with orthonormal columns, and a $p \times p$ real diagonal matrix $S = (s_{ij})$ whose diagonal entries are positive, such that $A = USV^*$ and $s_{ii} \geq s_{i+1,i+1}$ for all $i = 1, \dots, p-1$.*

The diagonal entries s_{ii} of S are the nonzero singular values of A . The columns of U span the column space of the matrix A ; hence, they provide an orthonormal basis of the column space of A . This property of the SVD is used in section 4 for the computation of orthonormal basis functions.

2.5. The QR method for finding the spectrum of a matrix. There are several methods available for determining the spectrum of a complex matrix (see, for example, [7]). In this paper, the QR method, as described in [12], is used for computing the eigenvalues of a complex linear operator on a finite-dimensional space.

2.6. Least squares Newton's method. Newton's method is a well-known numerical technique for solving equations of the form $F(x) = 0$, where $F : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a continuously differentiable function of the form

$$(2.11) \quad F(x) = \begin{pmatrix} f_1(x) \\ \vdots \\ f_n(x) \end{pmatrix},$$

and $x = (x_1, \dots, x_m)^T$. The method uses the Jacobian matrix J of F , which is defined by the formula

$$(2.12) \quad J(x) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x) & \cdots & \frac{\partial f_1}{\partial x_m}(x) \\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1}(x) & \cdots & \frac{\partial f_n}{\partial x_m}(x) \end{pmatrix},$$

and its pseudoinverse, which is denoted by $J^\dagger(x)$. We illustrate the application of the method in the following theorem, under the constraints relevant to our paper. When $m \neq n$, the procedure is referred to as the *least squares Newton's method* [13].

THEOREM 2.4. *Suppose $F : \mathbb{R}^m \rightarrow \mathbb{R}^n$ ($m \geq n$) is a continuously differentiable function and $\xi \in \mathbb{R}^m$ satisfies the condition*

$$(2.13) \quad F(\xi) = 0.$$

Furthermore, suppose that $J(x)$ defined by (2.12) has full row rank for all x in an ε -neighborhood $n_\varepsilon(\xi) = \{x : \|x - \xi\| < \varepsilon\}$ of ξ . Let $(x_k)_{k \geq 0}$ be a sequence of vectors defined by the formula

$$(2.14) \quad x_{k+1} = x_k - J^\dagger(x_k)F(x_k),$$

and let $(h_n)_{n \geq 0}$ be the sequence of differences

$$(2.15) \quad h_n = x_n - \xi.$$

Then there exist $\tilde{\varepsilon} > 0$, $\delta > 0$, and a positive integer $N_{\tilde{\varepsilon}, \delta}$ such that if $\|x_0 - \xi\| < \tilde{\varepsilon}$, then the inequality

$$(2.16) \quad \|h_{n+1}\| \leq \delta \|h_n\|^2$$

is true for all $n \geq N_{\tilde{\varepsilon}, \delta}$.

2.7. Quadratures. Quadrature rules are expressions of the form

$$(2.17) \quad \sum_{i=1}^n w_i \varphi(x_i),$$

where the points $x_i \in \mathbb{R}^d$ and the coefficients $w_i \in \mathbb{R}$ are referred to as the *nodes* and *weights* of the quadrature, respectively. They serve as approximations to integrals of the form

$$(2.18) \quad \int_{\Omega} \varphi(x) w(x) dx,$$

where w is a positive weight function and $\Omega \subset \mathbb{R}^d$ is the region of integration.

Quadratures are typically chosen so that (2.17) is equal to the desired integral (2.18) for some set of functions, commonly polynomials of some fixed order. A quadrature which integrates exactly all polynomials of degree at most k and does not integrate exactly some polynomial of degree $k+1$ is said to have *order* k . For instance, in one dimension, Gaussian quadrature rules consist of n nodes and have order $2n-1$.

Given a weight function w , an integration region Ω and m functions defined on Ω , a classical problem in numerical integration is to determine the minimum number of nodes n needed to integrate exactly all m functions and to find quadratures which achieve that minimum. Such quadratures are called *optimal*. In one dimension, Gaussian quadratures are known to be optimal rules for integrating polynomials up to a fixed degree. In higher dimensions, however, the theory is not complete [3].

Since each node $x \in \mathbb{R}^d$ is determined by d parameters and one additional parameter is given by the weight corresponding to the node, an n -point quadrature in $\mathbb{R}^d$ is determined by $n(d+1)$ parameters. For this reason, it is natural to expect that an optimal quadrature in d dimensions which integrates m functions has $n = \lceil \frac{m}{d+1} \rceil$ nodes, where $\lceil x \rceil$ denotes the smallest integer greater or equal to x . Based on this observation, the authors of [13] introduce an *efficiency* coefficient E to differentiate quadratures which integrate the same set of functions and have a different number of nodes. If n is the number of nodes of the quadrature, m is the number of functions which are integrated exactly, and d is the dimension of the Euclidean space, then the coefficient E is defined as

$$(2.19) \quad E = \frac{m}{n(d+1)}.$$

For Gaussian quadratures in one dimension, E is equal to 1. Following [13], we will refer to quadratures with E close to 1 as *efficient* quadratures.

2.8. Tensor product rules. An obvious way of obtaining quadratures in two dimensions is to combine two one-dimensional quadratures in the following way.

Let R be an m -point quadrature over an interval I ,

$$(2.20) \quad R(f) = \sum_{i=1}^m w_i f(x_i) \approx \int_I f(x) dx,$$

and S be an n -point quadrature over an interval J ,

$$(2.21) \quad S(f) = \sum_{j=1}^n v_j f(y_j) \approx \int_J f(y) dy.$$

The *tensor product* of R and S is the mn -point quadrature rule over the rectangle $I \times J$ defined by

$$(2.22) \quad R \times S(f) = \sum_{i=1}^m \sum_{j=1}^n w_i v_j f(x_i, y_j) \approx \int_{I \times J} f(x, y) dx dy.$$

The following observation is a simple consequence of the classical Fubini's theorem; a proof can be found, for instance, in [5].

THEOREM 2.5. *Let R be a quadrature rule that integrates $f(x)$ exactly over the interval I and let S be a quadrature rule that integrates $g(y)$ exactly over the interval J . Furthermore, let $h(x, y) = f(x)g(y)$. Then the tensor product quadrature $R \times S$ integrates $h(x, y)$ exactly over $I \times J$.*

Tensor product quadratures are relatively easy to construct; however, they are not optimal in terms of efficiency. For instance, let us consider two n -point Gaussian quadratures on the interval $[-1, 1]$, each of order $2n - 1$. Their tensor product is an n^2 -point quadrature on the square $[-1, 1] \times [-1, 1]$ of order $2n - 1$. (All monomials $x^i y^j$ of total degree $i + j$ at most $2n - 1$ can be expressed as products of monomials in the variables x and y of degree at most $2n - 1$; therefore, they are integrated exactly by the tensor product. However, there are polynomials of degree $2n$, for instance, x^{2n} , which are not integrated exactly by this quadrature.) The number of monomials $x^i y^j$ of total degree at most $2n - 1$ is $n(2n + 1)$; therefore, the efficiency coefficient of the tensor product of two Gaussian quadratures is

$$(2.23) \quad E = \frac{n(2n + 1)}{3n^2} = \frac{2n + 1}{3n},$$

a quantity which approaches $\frac{2}{3}$ as n increases.

3. Mathematical apparatus. The principal result of this section is Theorem 3.4, which allows us to discretize bounded convex regions in the plane. It is based on the observation that all eigenvalues of a certain linear operator on a finite-dimensional subspace of $L^2(X)$ fall within X , provided X is a convex, closed, and bounded set in the plane.

Our proof of Theorem 3.4 uses a geometrical observation introduced in the following section.

3.1. Three geometrical lemmas. This section consists of three geometrical observations: Lemma 3.1, which is a purely technical result employed in the proof of Lemma 3.2; Lemma 3.2, which is utilized to prove Lemma 3.3; and Lemma 3.3, which is used in the proof of Theorem 3.4.

LEMMA 3.1. *Let r be a positive real number and S be the square with vertices $(\pm r, \pm r)$ in the complex plane. Let $\lambda = x + iy$ be a point in the plane such that $x > r$ and μ be the point $(-\frac{xr}{x-r}, 0) = -\frac{xr}{x-r} + 0i$. Then μ lies outside S and*

$$(3.1) \quad |\lambda - \mu| > |z - \mu|$$

for all z in S .

Proof. First, we show that μ lies outside S . Since r is positive and x is greater than r , we have $0 < x - r < x$. Dividing the second inequality by $x - r$ yields

$$(3.2) \quad \frac{x}{x - r} > 1.$$

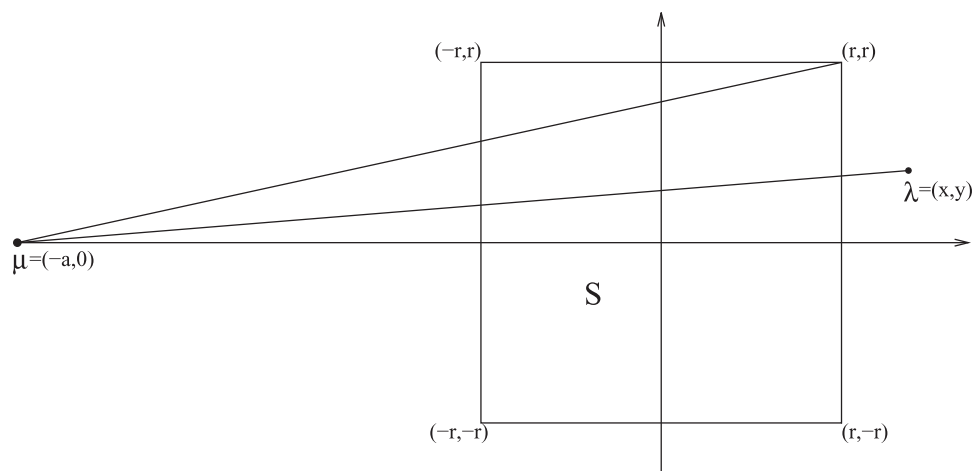


FIG. 3.1. The maximum distance from μ to a point in the square S is attained for one of the vertices $(r, \pm r)$; this distance is smaller than the distance from μ to λ .

Multiplying both sides of (3.2) by the negative number $-r$, we obtain

$$(3.3) \quad \frac{-xr}{x-r} < -r.$$

Inequality (3.3) says that the x -coordinate of μ is smaller than the x -coordinate of any point in the square S ; hence, μ lies outside S .

Second, we show that (3.1) holds for all z in S . Let a denote the quantity

$$(3.4) \quad a = \frac{xr}{x-r}$$

so that $\mu = (-a, 0) = -a + 0i$.

Obviously, the maximum distance from μ to a point in the square S is attained for the vertices $(r, \pm r)$ (see Figure 3.1); therefore,

$$(3.5) \quad \max_{z \in S} |z - \mu|^2 = |(r + ir) - (-a)|^2 = |r + a + ir|^2$$

$$(3.6) \quad = r^2 + 2ar + a^2 + r^2 = 2r^2 + 2ar + a^2.$$

On the other hand,

$$(3.7) \quad |\lambda - \mu|^2 = |(x + iy) - (-a)|^2 = x^2 + 2ax + a^2 + y^2.$$

We would like to prove that

$$(3.8) \quad |\lambda - \mu|^2 > \max_{z \in S} |z - \mu|^2.$$

To this end, we consider the quantity $|\lambda - \mu|^2 - \max_{z \in S} |z - \mu|^2$ and evaluate it using (3.6), (3.7), and (3.4):

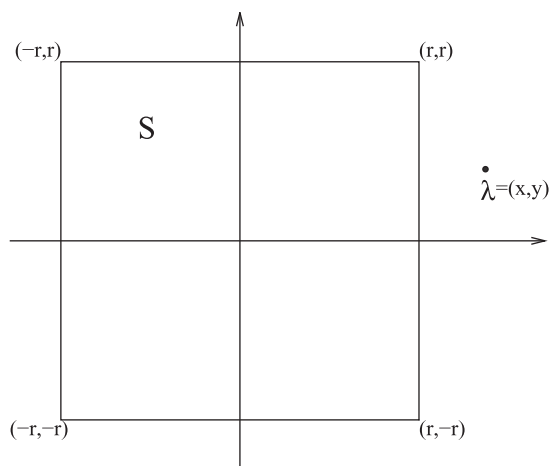


FIG. 3.2. The square S centered at the origin, with vertices $(\pm r, \pm r)$, and the point $\lambda = x + iy$ with $x > r$.

$$(3.9) \quad |\lambda - \mu|^2 - \max_{z \in S} |z - \mu|^2 = x^2 + 2ax + a^2 + y^2 - (2r^2 + 2ar + a^2)$$

$$(3.10) \quad = x^2 + 2ax + y^2 - 2r^2 - 2ar$$

$$(3.11) \quad = x^2 + y^2 - 2r^2 + 2a(x - r)$$

$$(3.12) \quad = x^2 + y^2 - 2r^2 + 2 \frac{xr}{x-r} (x - r)$$

$$(3.13) \quad = x^2 + y^2 - 2r^2 + 2xr$$

$$(3.14) \quad = x^2 + y^2 + 2r(x - r).$$

Since r is positive and x is greater than r , the right-hand side of (3.14) is a sum of three nonnegative terms, at least two of which are positive; therefore, the quantity $|\lambda - \mu|^2 - \max_{z \in S} |z - \mu|^2$ is positive. It follows that inequality (3.8) holds; taking square roots of both sides of (3.8) completes the proof. $\square$

LEMMA 3.2. Let S be a closed square region in the complex plane and $\lambda \in \mathbb{C}$ be a point outside S . Then there exists a point $\mu \in \mathbb{C}$ such that μ lies outside S and

$$(3.15) \quad |\lambda - \mu| > |z - \mu|$$

for all z in S .

Proof. By applying a rotation, we may assume that the sides of S are parallel to the coordinate axes. Furthermore, by applying a translation, we may assume that the center of S is at the origin. It follows that the vertices of S are located at the points $(\pm r, \pm r)$, where $2r$ is the length of the side of the square S .

Let $\lambda = x + iy$ with $x, y \in \mathbb{R}$. Since λ is not in S , at least one of the following inequalities holds: $|x| > r$ or $|y| > r$. By applying a rotation of 90, 180, or 270 degrees (which does not change the position of the square), we can ensure that

$$(3.16) \quad x > r \quad (\text{see Figure 3.2}).$$

Let $\mu = (-\frac{xr}{x-r}, 0)$. Obviously, the square S and the points λ and μ verify the hypothesis of Lemma 3.1; therefore, we can apply Lemma 3.1 to conclude that μ lies outside S and (3.15) holds for all z in S . $\square$

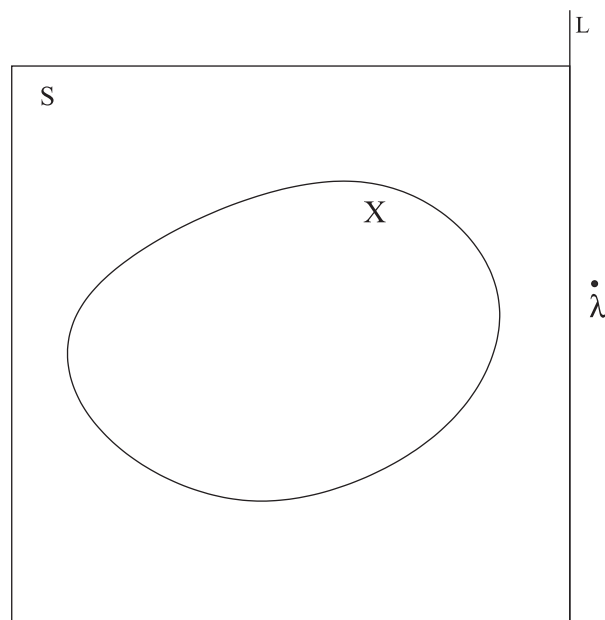


FIG. 3.3. X is enclosed in a square box S such that one of the sides of S lies along the line L , which separates the region X and the point λ .

LEMMA 3.3. *Let X be a convex, closed, and bounded subset of the complex plane and $\lambda \in \mathbb{C}$ be a point outside X . Then there exists a point $\mu \in \mathbb{C}$ such that μ lies outside X and*

$$(3.17) \quad |\lambda - \mu| > |z - \mu|$$

for all z in X .

Proof. By Lemma 2.2, there exists a line L which strictly separates X and λ . Since X is bounded, X can be enclosed in a square S ; moreover, S can be chosen in such a way that one of its sides lies along the line L (see Figure 3.3).

Since L separates λ and X , the square S does not contain λ . Consequently, we can apply Lemma 3.2 to conclude that there exists $\mu \in \mathbb{C}$ such that μ lies outside S and

$$(3.18) \quad |\lambda - \mu| > |z - \mu|$$

for all z in S . Since S contains X , it follows that μ lies outside X and

$$(3.19) \quad |\lambda - \mu| > |z - \mu|$$

for all z in X . $\square$

3.2. An analytical theorem. The principal analytical tool of this paper is the following theorem.

THEOREM 3.4. *Let X be a convex, closed, and bounded subset of $\mathbb{C}$, and let U be a finite-dimensional linear subspace of $L^2(X)$. Let M_{x+iy} be the multiplication operator defined by (2.2), and let P_U be the orthogonal projection onto the space U . Then all eigenvalues of the operator $P_U \circ M_{x+iy}$ fall inside X .*

Proof. The theorem is proved by contradiction. To this end, suppose the operator $P_U \circ M_{x+iy}$ has an eigenvalue $\lambda \in \mathbb{C}$ outside X . By Lemma 3.3, there exists $\mu \in \mathbb{C}$ such that $|\lambda - \mu| > |z - \mu|$ for all z in the closed and bounded set X ; hence,

$$(3.20) \quad |\lambda - \mu| > \max_{z \in X} |z - \mu|.$$

Since λ is an eigenvalue of the operator $P_U \circ M_{x+iy}$, there exists a nonzero $\varphi \in U$ such that

$$(3.21) \quad P_U(M_{x+iy}(\varphi)) = \lambda\varphi.$$

Subtracting $\mu\varphi$ from both sides of (3.21) we obtain

$$(3.22) \quad P_U(M_{x+iy}(\varphi)) - \mu\varphi = \lambda\varphi - \mu\varphi.$$

Since φ belongs to the subspace U and μ is a scalar, $\mu\varphi$ also belongs to U . By Theorem 2.1, it follows that $P_U(\mu\varphi) = \mu\varphi$; replacing $\mu\varphi$ by $P_U(\mu\varphi)$ in the left-hand side of (3.22) yields

$$(3.23) \quad P_U(M_{x+iy}(\varphi)) - P_U(\mu\varphi) = \lambda\varphi - \mu\varphi,$$

which is equivalent to

$$(3.24) \quad P_U(M_{x+iy}(\varphi) - \mu\varphi) = (\lambda - \mu)\varphi.$$

Next, we show that the L^2 -norm of the left-hand side of (3.24) is smaller than the L^2 -norm of the right-hand side of (3.24), which leads to a contradiction.

Let $\|\cdot\|$ denote the L^2 -norm on $L^2(X)$. By Theorem 2.1, the operator norm of the projection P_U is 1; therefore,

$$(3.25) \quad \|P_U(\psi)\| \leq \|\psi\|$$

for all ψ in $L^2(X)$. In particular, for $\psi = M_{x+iy}(\varphi) - \mu\varphi$, we have

$$(3.26) \quad \|P_U(M_{x+iy}(\varphi) - \mu\varphi)\| \leq \|M_{x+iy}(\varphi) - \mu\varphi\|.$$

Using the definition of the multiplication operator M_{x+iy} , (2.2b), we can express the right-hand side of (3.26) in the form

$$(3.27) \quad \|M_{x+iy}(\varphi) - \mu\varphi\| = \|(x + iy)\varphi - \mu\varphi\|$$

$$(3.28) \quad = \|(x + iy - \mu)\varphi\|.$$

The complex variable $x + iy$ takes values only within the set X ; therefore,

$$(3.29) \quad \|(x + iy - \mu)\varphi\| \leq \left(\max_{z \in X} |z - \mu| \right) \cdot \|\varphi\|.$$

Since φ is nonzero, $\|\varphi\|$ is positive; multiplying both sides of (3.20) by $\|\varphi\|$ yields

$$(3.30) \quad \left(\max_{z \in X} |z - \mu| \right) \cdot \|\varphi\| < |\lambda - \mu| \cdot \|\varphi\|.$$

Putting together (3.26), (3.27), (3.28), (3.29), and (3.30), we obtain

$$(3.31) \quad \|P_U(M_{x+iy}(\varphi) - \mu\varphi)\| < |\lambda - \mu| \cdot \|\varphi\|.$$

Since $\lambda - \mu$ is a scalar,

$$(3.32) \quad \|(\lambda - \mu)\varphi\| = |\lambda - \mu| \cdot \|\varphi\|.$$

Combining (3.31) with (3.32) yields

$$(3.33) \quad \|P_U(M_{x+iy}(\varphi) - \mu\varphi)\| < \|(\lambda - \mu)\varphi\|.$$

The last inequality says that the norm of the left-hand side of (3.24) is less than the norm of the right-hand side of (3.24); therefore, taking norms of both sides of (3.24) leads to the desired contradiction. $\square$

3.3. Remarks.

Remark 3.5. The eigenvalues of the operator $P_U \circ M_{x+iy}$ are complex numbers that fall inside X . Thus, they can be viewed as the nodes of a discretization of the region X that encapsulates information about the finite-dimensional space U . It follows that Theorem 3.4 provides a method for generating discretizations of bounded convex regions in two dimensions. For a given region, one can vary the choice of the subspace U in order to obtain discretizations suitable for different classes of functions.

Remark 3.6. Let G be the group of symmetries of X , i.e., G consists of all isometries under which X is invariant. The group G acts on the space $L^2(X)$ in the following way. For each $g \in G$ and $\varphi \in L^2(X)$, the action of g on φ is defined by

$$(3.34) \quad g(\varphi) = \varphi_g \in L^2(X),$$

$$(3.35) \quad \varphi_g(x, y) = \varphi(g^{-1}(x, y)) \quad \text{for all } (x, y) \in X.$$

Suppose the subspace U of $L^2(X)$ is invariant under the action of the group G , i.e.,

$$(3.36) \quad g(\varphi) \in U$$

for all $g \in G$ and $\varphi \in U$. Then the spectrum of the operator $P_U \circ M_{x+iy}$ is also invariant under the action of G , i.e., if λ is an eigenvalue of $P_U \circ M_{x+iy}$, then $g(\lambda)$ is also an eigenvalue of $P_U \circ M_{x+iy}$ for every symmetry g in G .

In other words, if the subspace U is invariant under the action of the symmetry group of X , the discretization given by Theorem 3.4 is symmetric.

Remark 3.7. Let $\{\varphi_1, \varphi_2, \dots, \varphi_n\}$ be an orthonormal basis of U . The eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of the operator $P_U \circ M_{x+iy}$ can be viewed as the nodes of an interpolation scheme on X relative to the functions $\{\varphi_1, \varphi_2, \dots, \varphi_n\}$. We observed that, for many choices of the region X and the space U , the condition number of the associated interpolation matrix (2.10) is small; therefore, the interpolation scheme is stable. In such cases, the eigenvalues can be used to obtain efficient quadrature rules on X in the following way.

Let $\{\psi_1, \psi_2, \dots, \psi_m\}$ ($m \geq n$) be a collection of functions on X which are to be integrated numerically. Usually, the functions ψ_i are selected such that the orthonormal basis $\{\varphi_1, \varphi_2, \dots, \varphi_n\}$ is a subset of $\{\psi_1, \psi_2, \dots, \psi_m\}$. Let

$$(3.37) \quad I_j = \iint_X \psi_j(x, y) dx dy, \quad j = 1, 2, \dots, m.$$

Constructing an n -point quadrature on X for the functions $\{\psi_j\}$ is equivalent to finding a zero of the function $F : \mathbb{R}^{3n} \rightarrow \mathbb{R}^m$,

$$(3.38) \quad F(x_1, \dots, x_n, y_1, \dots, y_n, w_1, \dots, w_n) = \begin{pmatrix} \sum_{i=1}^n \psi_1(x_i, y_i) w_i - I_1 \\ \dots \\ \sum_{i=1}^n \psi_m(x_i, y_i) w_i - I_m \end{pmatrix}.$$

We can find a zero of F using the least squares Newton's method provided that we have a good initial point, or, in other words, a good initial quadrature, to bootstrap Newton's method. It turns out that the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, viewed as nodes, together with suitably chosen weights, yield such an initial quadrature. Indeed, let $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ be defined by the equations

$$(3.39) \quad x_j + iy_j = \lambda_j, \quad j = 1, 2, \dots, n,$$

and let $(w_1, w_2, \dots, w_n)$ be the solution in the least squares sense of the linear system

$$(3.40) \quad \psi_j(x_1, y_1)w_1 + \psi_j(x_2, y_2)w_2 + \dots + \psi_j(x_n, y_n)w_n = I_j, \quad j = 1, 2, \dots, m.$$

It was observed that the least squares Newton's method, with initial point

$$(3.41) \quad (x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n, w_1, w_2, \dots, w_n)$$

defined by (3.39) and (3.40), returns a zero of the function F defined by (3.38), provided that $3n \geq m$ and the interpolation scheme given by the eigenvalues is stable. Moreover, the zero of F typically yields a quadrature with interior nodes and positive weights.

Finally, we note that the least squares Newton's method is very sensitive to the choice of the initial point. In our experience, among the various possible choices of nodes and weights of the initial quadrature, the best results (quadratures of high orders, with interior nodes and positive weights) were obtained when the initial quadrature was defined by (3.39) and (3.40).

4. Numerical algorithm. This section describes a numerical algorithm for the construction of a class of interpolation and quadrature formulae on convex regions in the plane.

Let X be a closed and bounded convex region in the plane. For any nonnegative integer n , we denote by $\mathcal{P}_n$ the space of polynomials on X of total degree at most n :

$$(4.1) \quad \mathcal{P}_n = \{P(x, y) : \text{total degree of } P \leq n\} \subset L^2(X).$$

One can easily see that the dimension of $\mathcal{P}_n$ is given by the formula

$$(4.2) \quad \dim \mathcal{P}_n = (n+1)(n+2)/2.$$

The algorithm makes use of Theorem 3.4, with $U = \mathcal{P}_n$, $P_U : L^2(X) \rightarrow U$ the orthogonal projection operator and M_{x+iy} the multiplication by $(x+iy)$ operator (as defined by (2.2)), to discretize the region X . The nodes of the discretization are used as initial nodes for the least squares Newton's method in order to construct quadrature rules on X .

The algorithm's input is

- X = the region to be discretized,
- n = the maximum degree of the polynomials in the space $U = \mathcal{P}_n$,
- m = the maximum degree of the polynomials to be integrated ($m \geq n$).

Let N and M denote the dimensions of $\mathcal{P}_n$ and $\mathcal{P}_m$, respectively. According to (4.2), $N = (n+1)(n+2)/2$ and $M = (m+1)(m+2)/2$.

The algorithm's output is a quadrature rule on X consisting of a set of N nodes $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$ and a set of N weights $w_1, w_2, \dots, w_N$ such that

$$(4.3) \quad \iint_X \varphi(x, y) dx dy \sim \sum_{j=1}^N \varphi(x_j, y_j) w_j$$

for any polynomial φ in the M -dimensional space $\mathcal{P}_m$.

The algorithm proceeds in four stages.

In the first stage, we construct an initial quadrature on X . In the second stage, we find an orthonormal basis of the space $\mathcal{P}_n$, iteratively, using the SVD. In the third stage, we find the eigenvalues of the operator $P_U \circ M_{x+iy}$. In the fourth stage we use the least squares Newton's method with step-length control to modify the position of the eigenvalues and of the initial weights (which are found using the method of least squares), in order to obtain a quadrature rule of order m on X .

Stage 1: Construct an initial quadrature on X . How the initial quadrature is constructed depends, of course, on the region X . In our implementation of the algorithm, only three well-known methods were used for constructing the initial quadrature: tensor product rules (described in section 2.8), decomposition into elementary regions, and change of variables. A survey of these and other methods of constructing quadratures on two-dimensional regions can be found in [5].

Stage 2: Construct an orthonormal basis of the space $\mathcal{P}_n$. In this stage, we construct an orthonormal basis of the space $\mathcal{P}_n$, defined by (4.1). The basis is used for determining the eigenvalues of the operator $P_U \circ M_{x+iy}$ on the space $\mathcal{P}_n$; therefore, the choice of the basis is not important—any basis yields the same eigenvalues. The following procedure for constructing an orthonormal basis is not optimal; however, it is stable and easy to implement.

(1) Consider the vector corresponding to the constant polynomial 1. Normalize it. The resulting vector provides an orthonormal basis of $\mathcal{P}_0$.

(2) Suppose we have constructed an orthonormal basis of $\mathcal{P}_k$, consisting of $K = (k+1)(k+2)/2$ vectors. Multiply each of the vectors in this basis by x and y , respectively, and put the resulting vectors, together with the vectors in the basis of $\mathcal{P}_k$, in a matrix having $3K$ columns. Calculate the SVD of the matrix. The SVD yields an orthonormal basis of the column space of the matrix, which is also an orthonormal basis of $\mathcal{P}_{k+1}$.

(3) Apply the procedure described in step (2) for each k in $\{0, 1, \dots, n-1\}$ in order to obtain an orthonormal basis of $\mathcal{P}_n$.

Stage 3: Find the eigenvalues of $P_U \circ M_{x+iy}$.

(1) Compute the entries (a_{ij}) of the matrix A representing $P_U \circ M_{x+iy}$ in the basis $\{\varphi_1, \varphi_2, \dots, \varphi_N\}$ constructed in Stage 2. To this end, use formula (2.9):

$$(4.4) \quad a_{ij} = \iint_X (x + iy) \varphi_i(x, y) \overline{\varphi_j(x, y)} dx dy;$$

the integrals in the right-hand side of (4.4) are evaluated using the quadrature constructed in Stage 1.

(2) Find the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_N$ of the complex matrix A using the QR algorithm described in section 2.5.

Stage 4: Construct the quadrature. In this stage, we construct an N -point quadrature of order m on X . We use the algorithm described in Remark 3.7.

(1) Construct an orthonormal basis $\psi_1, \psi_2, \dots, \psi_M$ of the space $\mathcal{P}_m$ using the procedure described in Stage 2. Find the integrals

$$(4.5) \quad I_j = \iint_X \psi_j(x, y) dx dy, \quad j = 1, 2, \dots, M,$$

using the quadrature constructed in Stage 1.

(2) The initial quadrature points $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$ for the least squares Newton's method are the eigenvalues

$$(4.6) \quad x_j + iy_j = \lambda_j, \quad j = 1, 2, \dots, N.$$

The initial quadrature weights $w_1, w_2, \dots, w_N$ are found by solving the following linear system using the method of least squares:

$$(4.7) \quad \psi_j(x_1, y_1)w_1 + \psi_j(x_2, y_2)w_2 + \dots + \psi_j(x_N, y_N)w_N = I_j, \quad j = 1, 2, \dots, M.$$

(3) Using the initial point

$$(4.8) \quad (x_1, x_2, \dots, x_N, y_1, y_2, \dots, y_N, w_1, w_2, \dots, w_N)$$

found in step (2), apply the least squares Newton's method with step-length control (see, for instance, [6] for a detailed discussion of the use of step-length control in Newton's method) to find a zero of the smooth function $F: \mathbb{R}^{3N} \rightarrow \mathbb{R}^M$ defined by

$$(4.9) \quad F(x_1, \dots, x_N, y_1, \dots, y_N, w_1, \dots, w_N) = \begin{pmatrix} \sum_{i=1}^N \psi_1(x_i, y_i)w_i - I_1 \\ \dots \\ \sum_{i=1}^N \psi_M(x_i, y_i)w_i - I_M \end{pmatrix}.$$

In the cases when the least squares Newton's method converges to a zero of F (cf. Remark 3.7), the zero defines an N -point quadrature on X which integrates all polynomials in the M -dimensional space $\mathcal{P}_m$.

Remark 4.1. We do not require the initial quadrature of X , constructed in Stage 1, to be efficient. The initial quadrature is used only for finding the eigenvalues and computing the integrals of the orthonormal basis polynomials; after these operations are performed, it can be discarded. Applying the least squares Newton's method, which is the most time-consuming part of the algorithm, does not make use of the initial quadrature.

Remark 4.2. Given a region X and a polynomial degree n , one can run the algorithm for various values of the order of the quadrature m . In order to obtain efficient quadratures, one runs the algorithm for increasing values of m until it fails to return quadratures with positive weights and interior nodes. In general, the largest value of m for which the algorithm does return a quadrature having positive weights

and interior nodes corresponds to a value of the parameter $M = (m+1)(m+2)/2$ approximately equal to $3N$. In other words, one can increase the degree m of the quadrature until the number of functions M to be integrated is approximately equal to three times the number $N = (n+1)(n+2)/2$ of nodes of the quadrature (cf. Remark 3.7). Therefore, for this largest value of the parameter m , the algorithm typically returns a quadrature with efficiency coefficient $E = \frac{M}{3N}$ close to 1 (see Tables 5.1 and 5.2).

In the following section we list the results of the application of the algorithm to triangles, regular polygons, and other convex regions.

5. Numerical results. We implemented the procedure described in section 4 and tested it on a number of examples. Several of the results are listed below in order to demonstrate the performance of the algorithm. The algorithm was implemented in Fortran and compiled with the Lahey–Fujitsu Fortran 95 compiler. Computations were performed in extended precision arithmetic on the triangle and in double precision arithmetic on all other regions.

5.1. Quadratures and interpolation formulae on the triangle. In this section we describe the quadratures and interpolation formulae on an equilateral triangle generated by the algorithm. For simplicity, we assume that the triangle is centered at the origin and has vertices located at the points $\{1, e^{2\pi i/3}, e^{4\pi i/3}\}$ on the unit circle.

In Tables 5.1 and 5.2 we list the parameters of the quadratures we obtained, together with the condition numbers of the associated interpolation matrices B (defined by (2.10)). The quadratures of orders 9, 11, 16, 21, 26, 29, and 31 have rotational symmetry, but not reflection symmetry, while all the remaining quadratures are fully symmetric—cf. Remark 5.1.

TABLE 5.1

Triangle quadratures, part 1. For each initial polynomial degree n from 1 to 10, we list the corresponding maximum order m of a quadrature having positive weights. The number of nodes of the quadrature is $N = (n+1)(n+2)/2$ and the number of polynomials which can be integrated exactly is $M = \dim \mathcal{P}_m = (m+1)(m+2)/2$. The efficiency of the quadrature is $E = M/(3N)$. The condition number of the associated interpolation matrix is listed in the row labeled CN .

n	1	2	3	4	5	6	7	8	9	10
m	2	4	5	7	9	11	12	14	16	17
N	3	6	10	15	21	28	36	45	55	66
M	6	15	21	36	55	78	91	120	153	171
E	0.67	0.83	0.70	0.80	0.87	0.93	0.84	0.89	0.93	0.86
CN	1.0	1.0	1.2	1.4	1.6	2.1	1.8	2.3	4.2	5.9

TABLE 5.2

Triangle quadratures, part 2. For each initial polynomial degree n from 11 to 19, we list the corresponding maximum order m of a quadrature having positive weights. The number of nodes of the quadrature is $N = (n+1)(n+2)/2$ and the number of polynomials which can be integrated exactly is $M = \dim \mathcal{P}_m = (m+1)(m+2)/2$. The efficiency of the quadrature is $E = M/(3N)$. The condition number of the associated interpolation matrix is listed in the row labeled CN .

n	11	12	13	14	15	16	17	18	19
m	19	21	22	23	26	27	29	31	32
N	78	91	105	120	136	153	171	190	210
M	210	253	276	300	378	406	465	528	561
E	0.90	0.93	0.88	0.83	0.93	0.88	0.91	0.93	0.89
CN	5.0	9.7	6.3	13.1	68.6	144.0	67.8	71.5	56.2

TABLE 5.3

A fully symmetric 45-point quadrature of order 14 on the triangle. Only 10 generating nodes are listed; the remaining nodes may be obtained via the action of the group of symmetries of the triangle.

x	y	Weight
-1.6868265317233866793E+00	0.29216692565001003106E+00	0.58935620335540765011E-01
-1.5808488336403957216E+00	-.12519480877035419803E-30	0.67452219369552326397E-01
-.35347157488571107334E+00	0.14203631281845219952E+00	0.44503540566573551812E-01
-.47009724117048709338E+00	0.41794071711619191254E-30	0.17897106324896028214E-01
-.36136906345529792533E+00	0.40253738576591622170E+00	0.36144179089957608755E-01
-.47259519998535818893E+00	0.24682585755277199100E+00	0.18293541893943411802E-01
-.47234427519430125315E+00	0.50374523862970667221E+00	0.16917794610623270344E-01
-.35112713454811437607E+00	0.60817003695340733938E+00	0.28921171272990801675E-01
-.47085943420511108264E+00	0.70900463999219136033E+00	0.12279549423359387150E-01
-.47949927002953977850E+00	0.83051709788335153417E+00	0.35293734203249423613E-02

TABLE 5.4

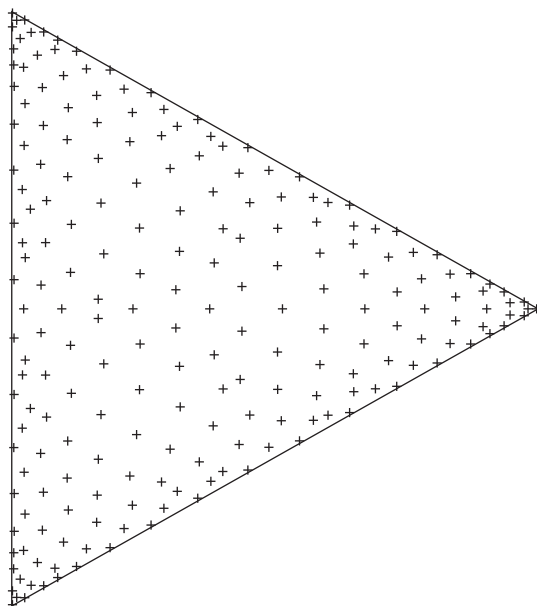
A fully symmetric 78-point quadrature of order 19 on the triangle. Only 16 generating nodes are listed; the remaining nodes may be obtained via the action of the group of symmetries of the triangle.

x	y	Weight
-.12399257021092653653E+00	0.19374825509442976738E-28	0.41450918407684045121E-01
-.13177700392715291090E+00	0.22824446607103231387E+00	0.38440468676165286217E-01
-.28778371647987152368E+00	0.10998294723245009451E+00	0.31317527185910258616E-01
-.40630672781840910415E+00	-.14456488256121493556E-28	0.14002753888728943368E-01
-.29724898552695112330E+00	0.31498865266126179066E+00	0.26827201883108899153E-01
-.41356124787667862272E+00	0.16719641141490991682E+00	0.19961899945473734702E-01
-.48264071535159024307E+00	0.95221898752783636336E-01	0.86539518883907294556E-02
-.41408886151213149256E+00	0.38336091989130268782E+00	0.20240988214359192248E-01
-.27842964300896413291E+00	0.48225428802479053639E+00	0.23989433658357479662E-01
-.48370152343347647465E+00	0.30142300793320924164E+00	0.87991644027906763838E-02
-.48281402069237875749E+00	0.49942229259718307221E+00	0.78178774263925103734E-02
-.40528714204595318336E+00	0.57394856816236337313E+00	0.18008780871442209707E-01
-.48105533677985120342E+00	0.64968083098649002499E+00	0.60973338938233607918E-02
-.41966365447253047169E+00	0.72687877163645268678E+00	0.10547116649857438346E-01
-.48630312842497788637E+00	0.76255697108968090949E+00	0.34615238341665281813E-02
-.48251983199382645129E+00	0.83574886467290610008E+00	0.22095115197099314446E-02

Remark 5.1. Since the eigenvalues are invariant under the action of the symmetry group of the triangle, in most cases, modifying their position by the least squares Newton's method yields quadratures on the triangle which are fully symmetric. In a few instances, however, relaxing the requirement that the quadratures have full symmetry, and allowing only rotational symmetry instead, allows us to obtain quadratures of higher orders m . The phenomenon has not been investigated further.

In Tables 5.3 and 5.4, we list the quadrature rules on the triangle of orders 14 and 19, respectively. Since each of the quadratures has full symmetry, only a subset of generating nodes and corresponding weights is sufficient for describing each of the quadratures. The remaining nodes may be obtained via the action of the symmetry group of the triangle, with the observation that points belonging to the same orbit have the same weight. (For a more detailed description, see [13].) The quadratures were computed in extended precision, integrate the orthonormal basis polynomials with 31 exact digits, and are listed with 20 digits.

Figure 5.1 shows a picture of the nodes of a fully symmetric quadrature of order 32.

FIG. 5.1. *Symmetric quadrature of order 32 on the triangle.*

5.2. Quadratures on regular polygons. Constructing quadratures on regular polygons having at least four sides is more involved. In contrast to the triangle case, the operator $P_U \circ M_{x+iy}$ has, in many instances, multiple eigenvalues; for all choices of the number of sides, the only multiple eigenvalue is 0, the center of the polygon.

We used the procedure described in the first three stages of the algorithm in section 4 to find the number of eigenvalues equal to 0 for all values of the number of sides k from 3 to 20 and all values of the maximum degree of polynomials n in the space $U = \mathcal{P}_n$ from 1 to 26. The results are listed in Table 5.5.

We note that the entries in the table seem to follow a certain pattern and, in fact, can be described by the following formulae.

CONJECTURE 5.2. *Let X be a regular polygon with k sides centered at 0, and let $U = \mathcal{P}_n$ be the space of polynomials on X of total degree at most n . We denote by $\Lambda(n, k)$ the number of eigenvalues equal to 0 of the operator $P_U \circ M_{x+iy}$ (as defined in the statement of Theorem 3.4) and by $\mathcal{S}_p$ the sum of the first p positive integers:*

$$(5.1) \quad \mathcal{S}_p = 1 + 2 + \cdots + p = p(p+1)/2$$

with the convention that $\mathcal{S}_{-1} = \mathcal{S}_0 = 0$. Then the following formulae hold.

If $k = 2t + 1$, where t is a positive integer,

$$(5.2) \quad \Lambda(n, k) = \begin{cases} \mathcal{S}_{n \pmod k + 1} & \text{if } n \pmod k \leq t - 1, \\ \mathcal{S}_{k - n \pmod k - 2} & \text{if } n \pmod k \geq t. \end{cases}$$

If $k = 2t$, where t is an integer greater or equal to 2,

$$(5.3) \quad \Lambda(n, k) = \begin{cases} \mathcal{S}_{n \pmod k + 1} + \frac{1}{2}(n - n \pmod k) & \text{if } n \pmod k \leq t - 1, \\ \mathcal{S}_{k - n \pmod k - 2} + \frac{1}{2}(n - n \pmod k + k) & \text{if } n \pmod k \geq t. \end{cases}$$

TABLE 5.5

The number of eigenvalues equal to zero for various regular polygons and spaces $\mathcal{P}_n$. The number of sides of the polygon is listed in the top row, while n is listed in the first column.

	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	0	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
2	0	2	1	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
3	1	2	0	4	3	10	10	10	10	10	10	10	10	10	10	10	10	10
4	0	3	0	3	1	7	6	15	15	15	15	15	15	15	15	15	15	15
5	0	5	1	3	0	5	3	11	10	21	21	21	21	21	21	21	21	21
6	1	4	3	4	0	4	1	8	6	16	15	28	28	28	28	28	28	28
7	0	4	1	6	1	4	0	6	3	12	10	22	21	36	36	36	36	36
8	0	5	0	9	3	5	0	5	1	9	6	17	15	29	28	45	45	45
9	1	7	0	7	6	7	1	5	0	7	3	13	10	23	21	37	36	55
10	0	6	1	6	3	10	3	6	0	6	1	10	6	18	15	30	28	46
11	0	6	3	6	1	14	6	8	1	6	0	8	3	14	10	24	21	38
12	1	7	1	7	0	11	10	11	3	7	0	7	1	11	6	19	15	31
13	0	9	0	9	0	9	6	15	6	9	1	7	0	9	3	15	10	25
14	0	8	0	12	1	8	3	20	10	12	3	8	0	8	1	12	6	20
15	1	8	1	10	3	8	1	16	15	16	6	10	1	8	0	10	3	16
16	0	9	3	9	6	9	0	13	10	21	10	13	3	9	0	9	1	13
17	0	11	1	9	3	11	0	11	6	27	15	17	6	11	1	9	0	11
18	1	10	0	10	1	14	1	10	3	22	21	22	10	14	3	10	0	10
19	0	10	0	12	0	18	3	10	1	18	15	28	15	18	6	12	1	10
20	0	11	1	15	0	15	6	11	0	15	10	35	21	23	10	15	3	11
21	1	13	3	13	1	13	10	13	0	13	6	29	28	29	15	19	6	13
22	0	12	1	12	3	12	6	16	1	12	3	24	21	36	21	24	10	16
23	0	12	0	12	6	12	3	20	3	12	1	20	15	44	28	30	15	20
24	1	13	0	13	3	13	1	25	6	13	0	17	10	37	36	37	21	25
25	0	15	1	15	1	15	0	21	10	15	0	15	6	31	28	45	28	31
26	0	14	3	18	0	18	0	18	15	18	1	14	3	26	21	54	36	38

It can be easily checked that the conjecture holds for all values of $\Lambda(n, k)$ listed in Table 5.5. A proof of the general case is beyond the scope of this paper.

Constructing quadratures in the cases when the operator $P_U \circ M_{x+iy}$ has multiple zero eigenvalues is more involved because the condition number of the interpolation matrix defined by (2.10) explodes. In several of these cases, Newton's method fails to find quadratures having interior nodes and positive weights. We illustrate the situation in Table 5.6. A dash denotes that no quadrature having positive weights was found, while a numbered entry denotes that quadratures having positive weights were found, and the numbered entry is the highest order of such a quadrature.

5.3. Quadratures on other regions. We used the procedure described in section 4 to construct quadratures on convex polygons, sectors of discs, ellipses, and other convex regions. A few sample quadratures are shown in Figures 5.2, 5.3, and 5.4.

The quadrature on the pentagon displayed in Figure 5.2 has 28 nodes and integrates 66 functions. The efficiency coefficient of the quadrature is 0.79 and the condition number of the associated interpolation matrix equals 30.5.

TABLE 5.6

Quadratures on regular polygons with at least four sides. The top row lists the initial maximum degree of polynomials n . A numbered entry denotes the maximum polynomial degree m for which a quadrature having positive weights was found. A dash denotes that no quadrature having positive weights was found.

	3	4	5	6	7	8	9	10
Square	5	5	-	9	-	11	-	13
Pentagon	4	7	6	9	10	12	11	
Hexagon	-	7	-	7	-	-	-	11
Heptagon	5	6	8	10	8	12	-	-
Octagon	-	-	6	9	9	13	-	-
Nonagon	-	-	8	10	11	13	14	14
Decagon	-	-	-	-	9	11	13	11

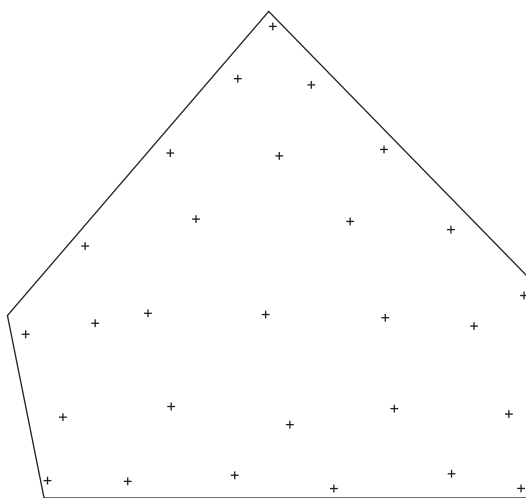


FIG. 5.2. A 28-point quadrature of order 10 on a convex pentagon.

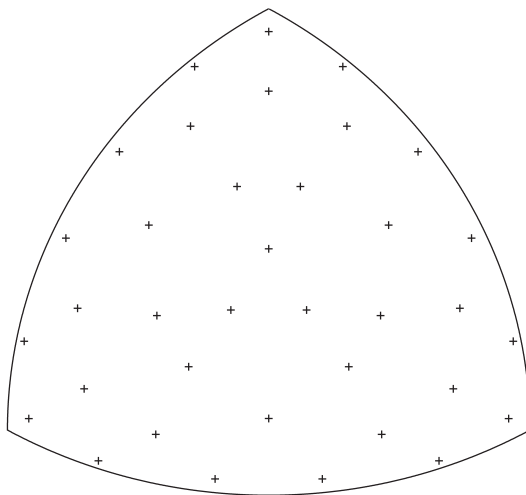


FIG. 5.3. A 36-point quadrature of order 11 on the Reuleaux triangle.

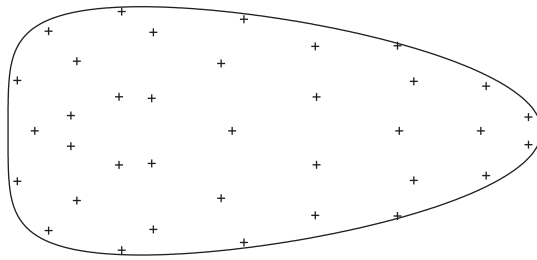


FIG. 5.4. A 36-point quadrature of order 12 on a convex region.

The quadrature on the Reuleaux triangle displayed in Figure 5.3 has 36 nodes and integrates 78 functions. The efficiency coefficient of the quadrature is 0.72 and the condition number of the associated interpolation matrix equals 88.3.

The quadrature on the convex region displayed in Figure 5.4 has 36 nodes and integrates 91 functions. The efficiency coefficient of the quadrature is 0.84 and the condition number of the associated interpolation matrix equals 27.1.

6. Conclusions. We have introduced a numerical algorithm for the construction of stable interpolation schemes and efficient quadratures on convex regions in the plane. The principal analytical tool used by the procedure is Theorem 3.4. The numerical examples illustrate the application of the theorem to triangles, regular polygons, and other convex regions. The quadratures constructed on these regions have interior nodes and positive weights and are invariant under the symmetry group of the region whenever such a nontrivial group is present. In addition, the quadratures exhibit Gaussian behavior—most of them have efficiency coefficients close to 1. The associated interpolation schemes are stable, as reflected by the low condition numbers of the corresponding interpolation matrices.

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